

Geometry Before Fields: Constraint-Reduced Cosmology and Optical Predictions in the Berger–Hopf Framework

Norman P. Carberry
Independent Researcher

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Abstract

We develop a geometry-first cosmological workflow in which physical degrees of freedom are identified only after action variation, constraint enforcement, gauge reduction, and physical-Hessian reduction, rather than assumed as fundamental fields. Applied to a closed hyperspherical parent geometry $S^3(R_H)$, the construction isolates a physical $n = 2$ reduced phase-space state and induces its metric response through the same constrained Hessian. The state is propagated to observations using a gauge-invariant closed- S^3 luminosity-distance kernel and Jacobi/Sachs optical transport for photon redshift, convergence, and shear. Unresolved local geometry is represented statistically, yielding the structural prediction rank $C_{\text{topo}} \leq m$ for m independent physical topographic texture channels; the one-channel limit is rank one. Effective-field/Horndeski variables are retained as compatibility languages rather than as the defining ontology. A curvature-dependent constraint-preservation defect in one effective representation, together with rejected preregistered foreground proxies, illustrates the distinction between representation and physical reduction. The framework produces falsifiable multi-observable tests while leaving the upstream action-owned source normalization explicitly open.

Keywords: cosmology; hyperspherical geometry; constrained dynamics; optical scalar equations; luminosity distance; weak lensing; stochastic geometry; falsifiable cosmology

1 Introduction

Late-time cosmological tensions and directional residuals motivate tests that are capable of separating local structure, survey geometry, and genuinely new large-scale dynamics. The scalar-topographic program introduced a compact hyperspherical parent $S^3(R_H)$ and a long-wavelength scalar response whose largest observational effect occurs at low redshift [1]. A later covariant formulation placed the topographic scalar in the cubic-Galileon sector of Horndeski theory, preserving second-order time evolution while allowing nontrivial late-time scalar–metric braiding.

The present work asks a narrower question: can the long-wavelength topographic response be represented as the large-scale branch of the same constraint-reduced response logic used in the Berger–Hopf program, without assuming that the full microscopic BHSM construction has already been compactified into cosmology? We answer this at the level of a controlled bridge model.

The central mathematical chain is

$$S[\Phi] \longrightarrow \hat{\mathcal{H}}_{\text{phys}} \longrightarrow \mathcal{K}_{\text{topo}} \longrightarrow \lambda_{\text{topo}} \longrightarrow T_n \longrightarrow \delta g_{\mu\nu} \longrightarrow \{D_L, H, f\sigma_8, \Phi + \Psi\}.$$

The distinction between established input and bridge extension is essential. The S^3 harmonic spectrum and cubic-Galileon background are treated as the cosmological starting point.

The Schur/Feshbach reduction is an exact mathematical operation once a physical complement is specified. The identification of its low-energy coefficients with the topographic response is a bridge assumption to be tested, not a declaration that the complete microscopic BHSM action has already been reduced to four-dimensional cosmology.

The paper makes four main contributions. First, it standardizes the S^3 harmonic convention and shows how the observed sky dipole arises from the four-dimensional $n = 1$ eigenspace rather than by identifying a radial mode with an angular dipole. Second, it derives a reduced kernel in which a positive higher-spatial-gradient term can emerge after integrating out a stable complement. Third, it identifies a parameter interval in which the $n = 1$ response is selectively softened while all $n \geq 2$ harmonics are stiffened. Fourth, it gives an executable preregistration protocol for curvature, supernova anisotropy, growth, and lensing tests.

The claim boundary is therefore:

This work does not assume that the scalar-topographic cosmological field has already been derived from the complete microscopic Berger–Hopf action. Exact reductions are distinguished from cross-scale identifications and exploratory reference-branch assumptions. Prospective observables are frozen before comparison with designated independent data.

2 Geometry before fields: the methodological thesis

The principal claim of this work is methodological before it is phenomenological. An effective field representation may be mathematically useful without uniquely identifying the underlying physical degrees of freedom. We therefore reverse the usual order of construction.

A representation-first workflow has the schematic form

$$\boxed{\text{choose fields} \rightarrow \text{write an effective action} \rightarrow \text{derive equations} \rightarrow \text{fit parameters} \rightarrow \text{compare with data.}} \quad (1)$$

The workflow pursued here is instead

$$\boxed{\text{geometry} \rightarrow \text{action} \rightarrow \text{constraints and boundary conditions} \rightarrow \text{gauge quotient} \rightarrow \text{physical tangent space} \rightarrow \text{physics}} \quad (2)$$

The distinction is operational, not semantic. In Eq. (2), the physical degrees of freedom are not decided before the constraint problem is solved. A scalar, vector, or other field-like variable that appears after reduction is therefore a coordinate on a surviving physical response unless an additional theorem establishes a more fundamental interpretation.

2.1 BHSM as a reduction program

BHSM implements this logic by constructing the constrained physical tangent space of a common variational problem and evaluating the action Hessian only after gauge and constraint reduction. If the physical variables are divided into a selected branch and its physical complement, the exact reduced response is obtained by the Schur/Feshbach map

$$\mathcal{K}_{\text{phys}} = A - BK^{-1}B^\dagger. \quad (3)$$

This identity is ordinary mathematics; its role here is to prevent nonphysical or redundant coordinates from being promoted to independent degrees of freedom.

At the cosmological scale the surviving physical topographic carrier is represented by the reduced phase-space state

$$X_2 = (q_2, \Pi_2)^T, \quad (4)$$

while the metric response is induced by the same physical Hessian,

$$h_2 = -\mathcal{A}_g^{-1} \mathcal{B}_{g_2} X_2. \quad (5)$$

The metric perturbation therefore carries no independent initial datum beyond the already retained physical mode.

2.2 Why cosmology is a useful stress test

Cosmology forces the construction to cross several logically distinct layers: compact geometry, harmonic selection, constraint reduction, metric response, matter response, photon propagation, and observer-frame projection. A representation that merely reproduces one layer is not automatically equivalent to the complete chain.

For this reason, the cosmological calculation serves as a stress test of the geometry-first methodology. The paper does not ask only whether an effective equation can be written. It asks whether one physical reduced state can be carried, without adding hidden degrees of freedom, from the action to observable quantities such as luminosity distance, growth, lensing, redshift perturbations, and their joint covariance.

2.3 Effective descriptions retain an important role

Nothing in this construction implies that effective field theory or quantum field theory is generally invalid. Effective descriptions are among the most successful tools in modern physics. The narrower point is that a useful effective representation should not automatically be identified with the fundamental ontology of the system.

Accordingly, effective scalar/Horndeski variables used in this work are retained as compatibility and translation languages. The defining transport law is the constraint-reduced physical response. Where an effective representation fails a curvature-dependent constraint-preservation audit, that failure is treated as evidence about the representation or its domain of validity rather than as permission to retune the underlying physical mode.

3 Claim status and falsifiability

Because this paper combines exact reduction statements, calibrated cosmological responses, failed candidate reductions, and still-open upstream source normalizations, we state their logical status explicitly.

Status	Statement	Role
Derived	The physical cosmological response is represented on a reduced two-dimensional $n = 2$ phase space, with metric response obtained from the same physical Hessian by block reduction.	Foundational transfer structure.
Derived	The closed- S^3 observer map includes the gauge-invariant luminosity-distance kernel and Jacobi/Sachs optical evolution for redshift, convergence, and shear.	Observable propagation.
Derived	For m independent physical topographic texture channels, $\text{rank } C_{\text{topo}} \leq m$ for the isolated topographic covariance.	Structural covariance test.
Calibrated / conditional	The numerical R1 cosmological realization and its absolute response normalization remain conditional on the retained-action map to the complete compact physical background.	Current phenomenological bridge.
Rejected	A homogeneous state-direction choice alone does not localize the response strongly enough in redshift.	Eliminated candidate mechanism.
Rejected	Frozen cumulative 2MRS number-density, luminosity-weighted, and BHSM seam-weighted foreground proxies do not transfer to the held-out sample.	Eliminated foreground reductions.
Open	The normalized compact full-preimage background, local action-owned metric/topographic matrix elements, exact seam-export normalization, and rare event statistics remain incomplete.	Upstream theory closure.

The distinction is essential to the methodology of this paper. A derived structural result is not promoted to a complete numerical prediction where an upstream normalization remains open, and a failed candidate is not repaired by post-hoc parameter changes.

3.1 What would falsify the present construction

The cosmological realization would be placed under direct tension if prospective data require, after standard covariance and noise treatment,

1. incompatible preferred angular axes among observables predicted to share the same one-mode response;
2. a redshift-transfer function inconsistent with the theory-selected physical state once that state is fixed independently;
3. topographic residual covariance of rank exceeding the number of physically admitted texture channels;
4. independent response degrees of freedom not supplied by the retained physical Hessian; or
5. a source normalization that cannot be obtained from the common retained action without introducing new phenomenological parameters.

These are failure conditions, not targets to be refit after observation.

4 Reproducibility and computational workflow

The computational results reported in this work are generated from versioned source code, machine-readable artifacts, frozen prediction manifests where applicable, and automated regression tests. Candidate mechanisms that failed their preregistered or held-out checks are retained as negative results rather than removed from the scientific record.

4.1 AI-assisted research tools

OpenAI ChatGPT and OpenAI Codex were used during the research program as interactive computational tools for symbolic-derivation assistance, software generation and revision, test orchestration, manuscript structuring, and interpretation of intermediate numerical results. Model versions varied over the development period and are recorded in project/session provenance where available. AI output was not accepted as evidence by itself: mathematical claims were transferred into explicit derivations, executable code, automated tests, frozen artifacts, or independently checkable numerical outputs before being retained. The author reviewed and edited all retained outputs and takes responsibility for the scientific content of this work.

The use of these tools does not change the claim hierarchy adopted throughout the paper. Results remain labeled as derived, calibrated/conditional, rejected, or open according to the underlying mathematical and empirical evidence.

5 Hyperspherical Parent and Harmonic Convention

We use a closed FLRW geometry

$$ds^2 = -dt^2 + a^2(t)R_H^2 \left[d\psi^2 + \sin^2 \psi d\Omega_2^2 \right],$$

with $a_0 = 1$. Define

$$D_C(z) = c \int_0^z \frac{dz'}{H(z')}, \quad \psi(z) = \frac{D_C(z)}{R_H}.$$

Scalar harmonics satisfy

$$-\nabla_\gamma^2 Y_n = \frac{n(n+2)}{R_H^2} Y_n, \quad n = 0, 1, 2, \dots,$$

so the physical spatial eigenvalue is

$$p_n(a) = \frac{n(n+2)}{a^2 R_H^2}.$$

A common closed-universe convention uses $k^2 = N^2 - 1$, hence

$$\boxed{N = n + 1}.$$

5.1 Exceptional first nonconstant harmonic

For our $n = 1$, $k^2 = 3$ and $N = 2$. The degree-one scalar harmonics on the unit three-sphere obey

$$D_i D_j Q^{(1)} = -\gamma_{ij} Q^{(1)}.$$

Therefore

$$\left(D_i D_j + \frac{k^2}{3} \gamma_{ij} \right) Q^{(1)} = 0.$$

The scalar-derived trace-free metric harmonic is absent. For the regular generally covariant one-scalar sector, the remaining scalar gauge transformations and lapse/shift constraints leave no ordinary propagating scalar degree of freedom in this exceptional sector. Modern gauge-invariant closed-universe analyses likewise take the propagating scalar spectrum to begin at alternative-convention $N \geq 3$ [2, 3].

Accordingly, the minimal bridge no longer uses our $n = 1$ as the physical scalar-metric carrier. An additional boundary degree of freedom could evade this conclusion, but it is not required for the construction below.

5.2 First physical scalar harmonic and exact dipole subspace

The first propagating scalar harmonic is

$$\boxed{n_\star = 2}, \quad \boxed{N_\star = 3}, \quad p_2 = \frac{8}{a^2 R_H^2}.$$

Let the unit $S^3 \subset \mathbb{R}^4$ embedding coordinates be

$$X^A = (\cos \psi, \sin \psi \hat{n}^i).$$

Every real degree-two scalar harmonic can be written

$$T_2 = A_{AB} X^A X^B, \quad A_{AB} = A_{BA}, \quad \delta^{AB} A_{AB} = 0.$$

Writing $c = \cos \psi$, $s = \sin \psi$, $a_i = A_{4i}$, and

$$A_{ij} = S_{ij} - \frac{A_{44}}{3} \delta_{ij}, \quad S_i^i = 0,$$

gives the exact observer-shell decomposition

$$\boxed{T_2 = A_{44} \left(c^2 - \frac{s^2}{3} \right) + 2cs \mathbf{a} \cdot \hat{\mathbf{n}} + s^2 S_{ij} \hat{n}^i \hat{n}^j.}$$

These terms are respectively a sky monopole, dipole, and quadrupole. The pure cross-term subspace

$$A_{44} = 0, \quad S_{ij} = 0, \quad a_i \neq 0$$

therefore gives an exact sky dipole:

$$\boxed{T_{2,\text{dip}} = A \sin(2\psi) (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}}).}$$

Thus the gauge audit shifts the carrier to the first physical compact-space scalar mode while retaining an exact dipolar observer-sky sector.

6 Constraint-Reduced Response and the Topographic Kernel

Let $\hat{\mathcal{H}}$ be the physical Hessian after constraints and gauge directions have been removed. For a selected topographic branch P and physical complement Q , the exact Schur/Feshbach reduced operator is

$$\mathcal{K} = P \left[\hat{\mathcal{H}} - \hat{\mathcal{H}} Q (Q \hat{\mathcal{H}} Q)^{-1} Q \hat{\mathcal{H}} \right] P.$$

For one scalar harmonic, write the quadratic block as

$$\hat{\mathcal{H}}_n = \begin{pmatrix} A(p) & C(p) \\ C^\dagger(p) & D(p) \end{pmatrix}.$$

The exact reduced stiffness is

$$K_{\text{red}}(p) = A(p) - C(p)D^{-1}(p)C^\dagger(p).$$

A minimal long-wavelength realization is

$$A(p) = Z_0 p, \quad D(p) = M_c^2 + c_c p, \quad C(p) = g\sqrt{p}.$$

Then

$$K_{\text{red}}(p) = Z_0 p - \frac{g^2 p}{M_c^2 + c_c p}.$$

For $c_c p \ll M_c^2$,

$$K_{\text{red}}(p) = \left(Z_0 - \frac{g^2}{M_c^2} \right) p + \frac{g^2 c_c}{M_c^4} p^2 + O(p^3).$$

Thus a positive p^2 stiffness can arise from a coupled second-order parent system after a stable complement is integrated out. In this interpretation the old static operator

$$-\nabla^2 + B\nabla^4$$

is a low-energy spatial derivative expansion, not evidence for a fundamental fourth-order time equation.

Define

$$\chi \equiv \frac{g^2}{M_c^2}, \quad L_c^2 \equiv \frac{c_c}{M_c^2}.$$

After canonical normalization of the leading spatial term, the reduced kernel may be written schematically as

$$K_{\text{topo}}(p) = m_T^2 + (c_s^2 - \chi)p + \chi L_c^2 p^2 + \dots.$$

The signs of the temporal kinetic term and of the full physical gradient matrix must be checked independently; a positive spatial p^2 coefficient by itself is not a complete ghost analysis.

7 Cubic-Galileon Scalar–Metric Bridge

The cosmological parent action is taken to be

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} \nabla_\mu T \nabla^\mu T + \frac{1}{\Lambda^3} \square T (\nabla T)^2 - V(T) + \mathcal{L}_m \right].$$

Using

$$X \equiv -\frac{1}{2} \nabla_\mu T \nabla^\mu T,$$

this corresponds to

$$G_2 = X - V(T), \quad G_3 = \frac{2X}{\Lambda^3}, \quad G_4 = \frac{M_{\text{Pl}}^2}{2}, \quad G_5 = 0$$

in the convention $\mathcal{L}_3 = -G_3 \square T$.

The model belongs to the kinetic-gravity-braiding subclass of Horndeski theory [4, 5]. Since G_4 is constant and $G_5 = 0$,

$$M_*^2 = M_{\text{Pl}}^2, \quad \alpha_M = 0, \quad \alpha_T = 0.$$

The scalar–metric mixing is controlled by the braiding function

$$\alpha_B = \frac{2\dot{T}^3}{HM_{\text{Pl}}^2\Lambda^3},$$

and

$$\alpha_K = \frac{\dot{T}^2}{H^2 M_{\text{Pl}}^2} + 6\alpha_B.$$

In the Bellini–Sawicki organization of Horndeski perturbations the scalar kinetic combination is

$$D_{\text{kin}} = \alpha_K + \frac{3}{2}\alpha_B^2.$$

Their perturbation equations are formulated for a spatially flat background [5]. We therefore use $D_{\text{kin}} > 0$ below as an inherited kinetic diagnostic for the cubic-Galileon sector, not as a complete proof of perturbative stability for the present closed background plus the additional reduced spatial response. A full stability proof requires the constrained gauge-invariant kinetic and gradient matrices, including the exceptional lowest scalar sector identified above.

For numerical work we softly break shift symmetry with

$$V(T) = V_0 - \mu^3 T.$$

The scalar energy density and pressure are

$$\rho_T = \frac{1}{2}\dot{T}^2 + V(T) + \frac{6H\dot{T}^3}{\Lambda^3},$$

$$P_T = \frac{1}{2}\dot{T}^2 - V(T) - \frac{2\dot{T}^2\ddot{T}}{\Lambda^3}.$$

The shift current

$$J = \dot{T} + \frac{6H\dot{T}^2}{\Lambda^3}$$

obeys

$$\dot{J} + 3HJ = \mu^3.$$

These background expressions are the standard kinetic-gravity-braiding relations specialized to the present G_2, G_3 choice [4].

The closed-FLRW background equation is

$$3M_{\text{Pl}}^2 \left(H^2 + \frac{1}{a^2 R_H^2} \right) = \rho_m + \rho_r + \rho_T.$$

The bridge does not require the bare Horndeski sound speed to become negative. Instead, the BHSM-inspired reduced response is used to lower the effective long-wavelength spatial stiffness while a positive p^2 term stiffens shorter wavelengths. This statement concerns the reduced spatial kernel; it is not, by itself, a complete closed-FLRW stability theorem.

8 BHSM-native transport of the physical topographic mode

The purpose of this section is to separate the ontology of the Berger–Hopf Standard Model (BHSM) mode from any particular four-dimensional effective-field-theory coordinate used to represent its late-time response. The distinction is important in the closed cosmological setting: the physical topographic carrier is not introduced as an additional free bulk scalar. Rather, it is an action-owned geometric/boundary mode whose physical dynamics are obtained only after the BHSM constraints and gauge directions have been removed.

8.1 Action ownership and physical tangent space

The BHSM construction used in this work treats modes as physical boundary-local fields acted on by sector projectors. The scalar/topographic sector is represented in the unified action candidate and admits an action-reduced mode functional together with a normalized profile-boundary boundary-value problem. These ingredients motivate the following physical tangent space at a background configuration $z_\star$:

$$T_{\text{phys}}(z_\star) = \ker DC(z_\star) \cap \mathcal{G}^\perp, \quad (6)$$

where $\mathcal{C}$ denotes the action-owned constraints and $\mathcal{G}$ the infinitesimal gauge directions. The corresponding projected Hessian is

$$\hat{\mathcal{H}} = P_{\text{phys}} D^2 \mathcal{L}_{\text{KKT}}(z_\star) P_{\text{phys}}. \quad (7)$$

If P_2 denotes the projector onto the first admissible physical scalar-topographic harmonic, the constraint-eliminated quadratic operator is the Schur–Feshbach reduction

$$\mathcal{K}_2 = P_2 \left[\hat{\mathcal{H}} - \hat{\mathcal{H}} Q_2 \left(Q_2 \hat{\mathcal{H}} Q_2 \right)^{-1} Q_2 \hat{\mathcal{H}} \right] P_2, \quad (8)$$

whenever the eliminated block is invertible on the chosen physical domain. Here $Q_2 = 1 - P_2$ on the already gauge-reduced tangent space.

8.2 Cosmological topographic-mode theorem

Theorem 1 (action-owned $n = 2$ envelope transport). Assume that:

1. the cosmological parent envelope is the closed spatial manifold $\mathcal{E}_{\text{cos}} \simeq S^3(R_H)$;
2. the scalar/topographic perturbation belongs to the BHSM action-owned boundary-mode sector and is reduced to T_{phys} before physical propagation is assigned;
3. the homogeneous $n = 0$ component is absorbed into the background and the exceptional scalar $n = 1$ sector is removed by the regular closed-FLRW constraint/gauge quotient;
4. the reduced $n = 2$ kinetic form is nondegenerate and positive on the retained mode.

Then the first physical scalar-topographic carrier is an action-owned $n = 2$ envelope mode. It does not require the introduction of an additional fundamental bulk scalar degree of freedom. Its canonical momentum, when nonzero, is induced by the reduced action itself and is therefore conjugate data of the same geometric mode rather than a separately postulated field.

Proof. Let $e_2(x)$ be a normalized element of $P_2 T_{\text{phys}}$ and write the retained perturbation locally as

$$\delta\Phi_{\text{topo}}(t, x) = q_2(t)e_2(x). \quad (9)$$

Substitution into the gauge- and constraint-reduced quadratic action gives, up to linear source terms and a choice of time parametrization,

$$S_2^{(2)} = \frac{1}{2} \int dt a^3 \left[\mathcal{G}_2(t) (D_t q_2)^2 - \mathcal{M}_2^2(t) q_2^2 + 2 \mathcal{J}_2(t) q_2 \right]. \quad (10)$$

The coefficients are the e_2 matrix elements of the reduced operator (8); explicitly, $\mathcal{G}_2$ is the kinetic quadratic form, $\mathcal{M}_2^2$ the reduced spatial/restoring form, and $\mathcal{J}_2$ the projection of the action-owned source onto e_2 . The canonical momentum is therefore

$$\Pi_2 = a^3 \mathcal{G}_2 D_t q_2, \quad (11)$$

which is generated by the same reduced action. Variation of (10) yields

$$D_t (a^3 \mathcal{G}_2 D_t q_2) + a^3 \mathcal{M}_2^2 q_2 = a^3 \mathcal{J}_2. \quad (12)$$

Thus the dynamical phase-space pair (q_2, Π_2) is the reduced description of one action-owned envelope mode. No second ontological scalar is created by introducing an effective coordinate for that mode. $\square$

8.3 Relation to the exact closed-Horndeski kernel

The closed-Horndeski calculation of Sec. 20 is used here as a late-time effective response representation of the same reduced mode, not as the microscopic definition of the BHSM topographic ontology. In the Bellini–Sawicki convention used in that section, the exact physical- $n = 2$ kinetic coefficient is

$$\mathcal{G}_{S,2} = M_{\text{Pl}}^2 \frac{5D_{\text{kin}}}{D_{\text{kin}} + 10(1 - \alpha_B^{\text{BS}}/2)^2}. \quad (13)$$

For $D_{\text{kin}} > 0$ and nonsingular braiding denominator this coefficient is positive. Equation (13) therefore supplies an effective kinetic matching condition for assumption 4 of Theorem 1 on the reference branch.

The Stückelberg variable π used in the EFT equations is accordingly treated as a coordinate representative of the retained response of q_2 . It is not counted as an additional BHSM mode. A change of EFT normalization amounts to an invertible local redefinition of the same reduced coordinate.

8.4 Exact observer dipole inside the physical $n = 2$ sector

A degree-two harmonic on S^3 may be written

$$T_2 = A_{AB} X^A X^B, \quad A_A{}^A = 0. \quad (14)$$

Relative to an observer shell,

$$T_2 = A_{44} \left(c^2 - \frac{s^2}{3} \right) + 2cs \mathbf{a} \cdot \hat{\mathbf{n}} + s^2 S_{ij} \hat{n}^i \hat{n}^j. \quad (15)$$

The pure A_{4i} subspace is therefore

$$\boxed{T_{2,\text{dip}} = A \sin(2\psi) (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}})}, \quad (16)$$

so the observed dipolar angular dependence is embedded exactly in the first physical scalar-topographic harmonic.

8.5 Role of the curved EFT Bianchi audit

The curvature-aware EFT system of the following sections remains a valuable cross-check of a chosen four-dimensional response representation. It is not used as the ontological definition of the BHSM mode and it is not allowed to introduce an additional independent scalar degree of freedom.

At the present stage, the literal curvature-aware 00, 0i, spatial-trace, and scalar equations and the reference background have been independently transcription-audited. The fully integrated curved-EFT differential- algebraic representation nevertheless retains a nonzero constraint- preservation defect, while the consistently flattened system closes to numerical precision. We therefore classify that calculation as an *EFT compatibility audit* rather than as the defining transport theorem.

The defining evolution statement for the bridge is Eq. (12): propagation is generated by the action-reduced physical $n = 2$ operator. A future exact identification of the curved EFT transfer map with $\mathcal{K}_2$ must reproduce this reduced phase space without duplicating degrees of freedom.

8.6 Claim boundary

Theorem 1 establishes the mode ownership and the form of its reduced transport law conditional on the BHSM action-owned scalar/topographic sector, the physical quotient, and positivity of the retained kinetic form. It does not by itself determine an absolute normalization for $\mathcal{M}_2^2$, $\mathcal{J}_2$, or the observable transfer coefficient. Those quantities are supplied, where stated, by the frozen reference branch or by explicit observational calibration and are labelled accordingly.

Consequently, the exact $n = 2$ dipole geometry, the physical-harmonic selection, and the positive effective kinetic matching are structural results. Reference growth curves and other quantities that require a particular EFT transfer realization remain reference-branch predictions rather than framework-wide theorems.

9 First numerical R1 physical- $n = 2$ reduced propagator

The representation-invariant prediction track can now be evaluated at the reduced gravity-scalar level. For the physical $n = 2$ harmonic, $-(\mathcal{D}^2 + 3K) = 5K$.

The exact reduced equation integrated here is

$$\boxed{\zeta_{,NN} + \left(3 + \frac{d \ln H}{dN} + \frac{d \ln \mathcal{G}_{S,2}}{dN}\right) \zeta_{,N} + \frac{5K}{a^2 H^2} \frac{\mathcal{F}_{S,2}}{\mathcal{G}_{S,2}} \zeta = 0.} \quad (17)$$

The fundamental matrix is normalized to the identity at $z = 2.1$ in the state $(\zeta, d\zeta/dN)^T$.

$$U(0, 2.1) = \begin{pmatrix} -0.186499277 & 0.019664322 \\ 0.013019584 & -0.005651284 \end{pmatrix}. \quad (18)$$

Across $0 \leq z \leq 2.1$ the numerical scan gives $\min \mathcal{G}_{S,2} = 1.471470e - 03$ and $\min \mathcal{F}_{S,2} = 5.037267e - 01$. The maximum absolute residual of the exact lapse/shift constraints over both

fundamental solutions is $2.720e - 14$, and the Liouville determinant identity is reproduced with relative residual $4.171e - 14$.

9.1 Claim boundary

This is the first numerical phase-space transfer kernel for the physical $n = 2$ R1 mode. It uses the exact reduced closed-Horndeski $n = 2$ kinetic coefficient and constraints. The current $\mathcal{F}_{S,2}$ implementation is explicitly the gravity-scalar expression; the complete matter+radiation scalar stability/response system is not yet included. Consequently this result is not yet the directional $f\sigma_8$, lensing, or luminosity-distance observable kernel.

10 Direct $n = 2$ matter-growth and Weyl transfer

The reduced physical $n = 2$ propagator can be coupled directly to minimally coupled matter through covariant energy-momentum conservation, without invoking the curvature-defective Newtonian-gauge DAE.

For pressureless matter in unitary gauge,

$$\dot{v}_m = -\delta n, \quad (19)$$

$$\dot{\delta}_m + 3\frac{d}{dt}(Hv_m) = -\left(3\dot{\zeta} + \frac{k_2^2}{a^2}\chi\right) + \frac{k_2^2}{a^2}v_m, \quad (20)$$

with $k_2^2/a^2 = 8K/a^2$ for the physical manuscript $n = 2$ harmonic.

In the same unitary-gauge convention the gauge-invariant potentials are

$$\Psi = \delta n + \dot{\chi}, \quad \Phi = \zeta + H\chi, \quad (21)$$

so the local Weyl/lensing combination in this sign convention is $\Psi - \Phi$.

The response is integrated for both basis columns of the physical state $(\zeta, d\zeta/dN)^T$, with zero matter response at the $z = 2.1$ anchor. Thus the result is a row-vector kernel on the same two-dimensional physical state space.

10.1 Present-epoch state-space kernels

$$\mathcal{F}_{f\sigma_8}^{\text{state}}(0) = (2.172303247e + 00, 3.071170118e - 02), \quad (22)$$

$$\mathcal{F}_{\text{Weyl}}^{\text{state}}(0) = (6.167895805e - 02, -2.498933170e - 02). \quad (23)$$

These are not two new amplitudes. They are the two components of linear functionals acting on the same initial $n = 2$ state. The single-calibration theorem collapses them to one observable prediction once the luminosity-distance light-cone functional fixes that state.

10.2 Claim boundary

The $f\sigma_8$ object above is a state-space fractional response kernel, using the R1 baseline dD/dN normalization. It is not yet the final single-number directional prediction because the full D_L calibration functional has not yet selected the initial-state vector. Likewise $\Psi - \Phi$ is the local Weyl potential combination; the integrated convergence/shear line-of-sight kernel remains to be evaluated.

11 Gauge-invariant luminosity-distance kernel and calibration-rank audit

The remaining light-cone observable is evaluated with the gauge-invariant geometric luminosity-distance construction.¹ For positive spatial curvature the background transverse distance and lensing efficiency use $S_K(\chi) = \sin(\sqrt{K}\chi)/\sqrt{K}$, with the corresponding curved-FLRW line-of-sight kernel.²

For the pure A_{4i} physical $n = 2$ mode, per unit $\mu = \hat{\mathbf{n}} \cdot \hat{\mathbf{p}}$,

$$Q_{2,\text{dip}}(\chi) = \sin(2\sqrt{K}\chi), \quad \hat{\nabla}^2 Q_{2,\text{dip}} = -2Q_{2,\text{dip}}. \quad (24)$$

The fixed-observed-redshift distance fluctuation is

$$\boxed{\frac{\delta D_L}{D_L} = \phi_s + \delta z + \frac{C_K(\chi_s)}{S_K(\chi_s)} \delta \chi - \kappa,} \quad (25)$$

with $C_K(\chi) = \cos(\sqrt{K}\chi)$ and the redshift, radial and convergence distortions evaluated along the same R1 physical- $n = 2$ metric response.

11.1 State-space calibration row

At the reference bin center $z_\star = 0.02$, the resulting luminosity-distance row on the $(\zeta, d\zeta/dN)$ state anchored at $z = 2.1$ is

$$\boxed{\mathcal{F}_{D_L}(z_\star) = (1.842472667e - 01, -3.520884651e - 03).} \quad (26)$$

The carried-forward SN calibration imposes one scalar constraint on this two-dimensional state:

$$\mathcal{F}_{D_L}(z_\star) \cdot X_{2,\text{anchor}} = -1.897330117e - 02 \pm 4.144653167e - 03. \quad (27)$$

The corresponding unit null direction is

$$n_{\text{cal}} = (1.910607424e - 02, 9.998174623e - 01), \quad \mathcal{F}_{D_L} \cdot n_{\text{cal}} = 0. \quad (28)$$

11.2 Calibration-rank consequence

This numerical light-cone calculation exposes a logical condition implicit in the single-calibration theorem: one amplitude calibration determines the model only after the normalized physical mode trajectory $\hat{X}_2$ has been fixed independently. The luminosity-distance row has rank one on a two-dimensional phase space. The growth and Weyl rows are not generally parallel to it, so motion along n_{cal} leaves the SN calibration unchanged while altering the other observables.

The next prediction-closure object is the theory-selected normalized state direction $\hat{X}_2$.

11.3 Claim boundary

No second observational calibration is introduced. Until the branch direction is fixed by the retained action or by an explicit preregistered theoretical branch condition, the manuscript reports the observable kernels as state-space rows rather than unique scalar amplitudes.

¹J. Yoo and F. Scaccabarozzi, JCAP 09 (2016) 046, arXiv:1606.08453.

²See also T. Pyne and M. Birkinshaw, MNRAS 348 (2004) 581 for the linear luminosity distance in curved perturbed FLRW spacetimes.

12 Local BHSM topography on the hyperspherical parent

The smooth $S^3(R_H)$ background and the physical $n = 2$ response do not exhaust the geometry sampled by a photon. We therefore separate the global parent from a local deformation field $\psi(\mathbf{x}, t)$:

$$\gamma_{ij} = a^2 e^{2\psi} \bar{\gamma}_{ij}^{S^3}. \quad (29)$$

To first order its intrinsic-curvature perturbation is

$$\delta^{(3)}R = -\frac{4}{a^2} \left(\bar{\nabla}^2 + \frac{3}{R_H^2} \right) \psi, \quad (30)$$

so the physical manuscript $n = 2$ harmonic carries the shifted factor $8 - 3 = 5$. The local volume rate is

$$H_{\text{local}} = H + \dot{\psi}. \quad (31)$$

The retained BHSM topographic flux candidate is $F_T = \text{abla}T - \text{Babla}(\text{abla}^2 T)$ with $\nabla \cdot F_T = S$. The existing seam-charge candidate is

$$V_\Sigma^2 = \frac{c^2}{8\pi} \int_{\Sigma_\star} H_\Sigma \Delta K_{uu} dA, \quad (32)$$

giving $V_\Sigma^2 = GM/r_\star$ at leading spherical weak-field order. The exact action-level export from this boundary charge into the dynamical ψ source remains open; no normalization is invented here.

12.1 Localized line-of-sight response

For a minimal coherent local-source proxy the accumulated radial perturbation grows only to a coherence distance χ_c and then saturates. The closed- S^3 radial distance term therefore has the normalized shape

$$\mathcal{F}_{\text{loc}}(z) = \frac{[C_K/S_K](\chi) \min(\chi, \chi_c)}{[C_K/S_K](\chi_\star) \min(\chi_\star, \chi_c)}. \quad (33)$$

The numerical example below uses $z_c = 0.03$ only as an illustrative local-coherence proxy, not as a fitted or frozen model parameter.

z	$A_\mu^{\text{loc}} [\text{mag}]$
0.02	-0.041200
0.10	-0.012598
0.30	-0.004435
0.50	-0.002810
0.80	-0.001904
1.00	-0.001605
1.50	-0.001212
2.10	-0.000991

Calibration boundary. The carried-forward low-redshift SN scalar amplitude normalizes the local topographic response in this exploratory decomposition. It cannot simultaneously determine an independent global $n = 2$ amplitude. The global/local split must ultimately be fixed by the common BHSM action or an independent preregistered observable.

13 Frozen 2MRS foreground-transfer test

The illustrative local-coherence cutoff is replaced here by a foreground field constructed from the 2MRS galaxy distribution. For supernova i ,

$$\mathcal{F}_i = \frac{C_K(\chi_i)}{S_K(\chi_i)} \sum_s \Delta\chi_{is} \delta_s(\hat{\mathbf{n}}_i),$$

where δ_s is the fixed 10° Gaussian-smoothed density contrast in shell s and $\Delta\chi_{is}$ is only the portion of that shell lying in front of the source. The score is standardized without using supernova residuals. Only one foreground amplitude is fitted.

The frozen analysis used 277 Pantheon+ Hubble-flow supernovae and 38600 2MRS galaxies with $z < 0.05$. Relative to the fixed-axis plus nuisance-control model, adding the foreground score gave

$$\Delta\chi^2 = 1.619359, \quad \beta_F = 0.015286 \pm 0.007846 \text{ mag},$$

with a redshift-blocked permutation value

$$p_{\text{perm}} = 0.076.$$

The preregistered classification is **not_established**.

As a transfer check, the foreground coefficient was calibrated only for $0.01 \leq z \leq 0.03$ and then held fixed for $0.03 < z \leq 0.15$. The held-out change was

$$\Delta\chi_{\text{heldout}}^2 = -4.107958.$$

This test does not identify the physical source of the foreground signal. The 2MRS field is a matter-density proxy, not yet an action-normalized BHSM de-encapsulation map, and conventional peculiar-velocity, lensing and survey selection effects remain alternative explanations.

14 BHSM optical transfer through a stochastic topographic geometry

The preceding sections separate the closed hyperspherical parent $S^3(R_H)$, the physical BHSM $n = 2$ carrier, and local line-of-sight topography. A realistic photon path, however, cannot be represented faithfully as a finite sum over foreground objects. In geometric optics the correct compression is the optical tidal matrix sampled by the null bundle. This section therefore derives a first-order BHSM optical transfer system in which unresolved foreground complexity is represented by a continuous geometric source and, when appropriate, by its path covariance.

14.1 Exact null-bundle geometry

Let k^μ be the tangent to the central null geodesic,

$$k^\nu \nabla_\nu k^\mu = 0, \quad k^\mu k_\mu = 0, \tag{34}$$

and let e_A^μ , $A = 1, 2$, be a parallel-transported Sachs screen basis. The Jacobi matrix $\mathcal{D}_{AB}$ obeys

$$\frac{d^2 \mathcal{D}}{d\lambda^2} = \mathcal{T} \mathcal{D}, \quad \mathcal{T}_{AB} = -R_{\mu\nu\alpha\beta} e_A^\mu k^\nu k^\alpha e_B^\beta. \tag{35}$$

The photon frequency for an observer congruence u^μ is

$$\omega = -u_\mu k^\mu, \quad \frac{d\omega}{d\lambda} = -k^\mu k^\nu \nabla_\mu u_\nu. \quad (36)$$

For the closed background define

$$S_K(\chi) = \frac{\sin(\sqrt{K}\chi)}{\sqrt{K}}, \quad C_K(\chi) = \cos(\sqrt{K}\chi), \quad H_K(\chi) = \frac{C_K(\chi)}{S_K(\chi)}. \quad (37)$$

Writing

$$\mathcal{D} = S_K(\chi)[I + \mathcal{A}] \quad (38)$$

and linearizing $\mathcal{T} = -KI + \delta\mathcal{T}$ gives

$$\boxed{\mathcal{A}'' + 2H_K\mathcal{A}' = \delta\mathcal{T}.} \quad (39)$$

14.2 Seven-state optical system

Decompose

$$\mathcal{A} = -\kappa I - \begin{pmatrix} \gamma_1 & \gamma_2 \\ \gamma_2 & -\gamma_1 \end{pmatrix} \quad (40)$$

and define

$$\mathcal{F} = -\frac{1}{2}\text{Tr } \delta\mathcal{T}, \quad \mathcal{G}_1 = -\frac{1}{2}(\delta\mathcal{T}_{11} - \delta\mathcal{T}_{22}), \quad \mathcal{G}_2 = -\delta\mathcal{T}_{12}. \quad (41)$$

Then

$$\kappa'' + 2H_K\kappa' = \mathcal{F}, \quad (42)$$

$$\gamma_1'' + 2H_K\gamma_1' = \mathcal{G}_1, \quad (43)$$

$$\gamma_2'' + 2H_K\gamma_2' = \mathcal{G}_2. \quad (44)$$

Let

$$\zeta_\gamma = \delta \ln(1+z), \quad \zeta'_\gamma = \mathcal{Z}(\chi), \quad (45)$$

and define

$$Y_\gamma = (\zeta_\gamma, \kappa, p_\kappa, \gamma_1, p_1, \gamma_2, p_2)^T. \quad (46)$$

The optical propagation law is

$$\boxed{\frac{dY_\gamma}{d\chi} = A_K Y_\gamma + B_{\text{opt}} S_{\text{opt}},} \quad (47)$$

where

$$S_{\text{opt}} = (\mathcal{Z}, \mathcal{F}, \mathcal{G}_1, \mathcal{G}_2)^T, \quad (48)$$

$$A_K = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -2H_K & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2H_K & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -2H_K \end{pmatrix}, \quad (49)$$

and

$$B_{\text{opt}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (50)$$

14.3 BHSM closure

The physical global mode is $X_2 = (q_2, \Pi_2)^T$ and the existing metric-response theorem gives

$$h_2 = \mathcal{R}_{g2} X_2, \quad \mathcal{R}_{g2} = -\mathcal{A}_g^{-1} \mathcal{B}_{g2}. \quad (51)$$

For local topography define

$$h_{\text{topo}} = \mathcal{R}_{g\psi} \psi, \quad \mathcal{R}_{g\psi} = -\mathcal{A}_g^{-1} \mathcal{B}_{g\psi}, \quad (52)$$

with its complete cosmological normalization left open.

The local field is written

$$\boxed{\mathcal{L}_\psi \psi = J_{\text{grav}} + \mathcal{E}_\Sigma[Q_\Sigma] + \xi,} \quad (53)$$

where the static spatial limit of $\mathcal{L}_\psi$ must reproduce $K_T = -\nabla^2 + B\nabla^4$ and

$$Q_\Sigma = V_\Sigma^2 = \frac{c^2}{8\pi} \int_{\Sigma_*} H_\Sigma \Delta K_{uu} dA. \quad (54)$$

The seam-export operator $\mathcal{E}_\Sigma$ is explicitly action-normalization-open.

If $\mathcal{O}_{\text{opt}}$ maps the physical metric response to $(\mathcal{Z}, \mathcal{F}, \mathcal{G}_1, \mathcal{G}_2)$ and $\mathcal{G}_\psi = \mathcal{L}_\psi^{-1}$, then

$$\boxed{\begin{aligned} \frac{dY_\gamma}{d\chi} &= A_K Y_\gamma \\ &+ B_{\text{opt}} \mathcal{O}_{\text{opt}} [\mathcal{R}_{g2} X_2 + \mathcal{R}_{g\psi} \mathcal{G}_\psi (J_{\text{grav}} + \mathcal{E}_\Sigma[Q_\Sigma] + \xi)]. \end{aligned}} \quad (55)$$

14.4 Stochastic unresolved terrain

Write

$$S_{\text{opt}} = \bar{S}_{\text{opt}} + \xi_{\text{opt}} \quad (56)$$

with

$$\langle \xi_{\text{opt}} \rangle = 0, \quad \langle \xi_{\text{opt}}(\chi) \xi_{\text{opt}}^T(\chi') \rangle = Q_{\text{opt}}(\chi) \delta(\chi - \chi'). \quad (57)$$

The path mean and covariance obey

$$\boxed{\bar{Y}'_\gamma = A_K \bar{Y}_\gamma + B_{\text{opt}} \bar{S}_{\text{opt}}} \quad (58)$$

and

$$\boxed{C'_Y = A_K C_Y + C_Y A_K^T + B_{\text{opt}} Q_{\text{opt}} B_{\text{opt}}^T.} \quad (59)$$

This is the computational compression used here: unresolved foreground complexity is carried by a 7×7 path covariance rather than by an exhaustive foreground-object ledger.

Rare nonlinear events may be represented separately by a compound-jump term,

$$dY_\gamma = (A_K Y_\gamma + B_{\text{opt}} \bar{S}_{\text{opt}}) d\chi + B_{\text{opt}} L dW_\chi + B_{\text{opt}} J dN_\chi. \quad (60)$$

14.5 Observable map and claim boundary

At a fixed physical source endpoint,

$$D_L = (1 + z)^2 D_A \quad (61)$$

gives

$$\delta \ln D_L = 2\zeta_\gamma - \kappa. \quad (62)$$

For a Hubble diagram at fixed observed redshift, the accumulated optical state is passed through the existing closed- S^3 source remapping,

$$\frac{\delta D_L}{D_L} = \phi_s + \delta z + \frac{C_K(\chi_s)}{S_K(\chi_s)} \delta \chi - \kappa. \quad (63)$$

Thus Eq. (55) supplies the accumulated physical path state; it does not replace the gauge-invariant luminosity-distance kernel.

The exact closed- S^3 optical propagator, the seven-state reduction and the covariance transport are derived here. The action-normalized seam export $\mathcal{E}_\Sigma$, the complete local metric-response map $\mathcal{R}_{g\psi}$, the stochastic covariance Q_{opt} and any rare-event jump law remain open constitutive objects. A future decisive test is therefore not an object-by-object mass reconstruction but a shared cross-observable covariance test: one optical path law must jointly predict luminosity-distance scatter, convergence/shear statistics, redshift perturbations and their cross-correlations.

15 Constitutive reduction of the BHSM optical source

The optical transfer equation of Sec. 14 isolates two constitutive maps that were left open: the seam-to-topography export $\mathcal{E}_\Sigma$ and the local metric-response map $\mathcal{R}_{g\psi}$. The retained BHSM calculations already reduce both maps further than a generic phenomenological parameterization.

15.1 Static and moving seam export

The retained scalar-topographic flux obeys

$$\mathbf{F}_T = \nabla T - B \nabla (\nabla^2 T), \quad \nabla \cdot \mathbf{F}_T = S, \quad (64)$$

and the canonical-flux acceleration candidate is

$$\mathbf{g}_T = -Q \mathbf{F}_T. \quad (65)$$

For the stationary spherical neighborhood,

$$Q S_{\text{enc}}(r) = \frac{Q_\Sigma}{r^2}, \quad Q_\Sigma = V_\Sigma^2, \quad r \geq r_\star. \quad (66)$$

Thus the static export profile is

$$\boxed{\mathcal{E}_\Sigma^{(0)}[Q_\Sigma] = \epsilon_\Sigma \frac{Q_\Sigma}{Q} \frac{\Theta(r - r_\star)}{r^2}.} \quad (67)$$

The r^{-2} shape is inherited from the conserved-throughput neighborhood law; ϵ_Σ records the still-open common-action orientation.

If de-encapsulation is represented by the time change of this exported source, then for a moving spherical seam

$$\mathcal{E}_\Sigma^{(1)} = \frac{\epsilon_\Sigma}{Q} \left[\dot{Q}_\Sigma \frac{\Theta(r - r_\star)}{r^2} - Q_\Sigma \dot{r}_\star \frac{\delta(r - r_\star)}{r_\star^2} \right]. \quad (68)$$

The second term is a seam-localized contribution generated by boundary motion. Equation (68) is a minimal kinematic continuation, not yet a proof that the retained action selects the time derivative rather than the static source itself.

15.2 Explicit local metric response

Choose a two-coordinate physical scalar metric basis $h_s = (\Phi, \Psi)^T$ and write

$$\mathcal{A}_g^{\text{loc}} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad \mathcal{B}_{g\psi}^{\text{loc}} = \begin{pmatrix} u \\ v \end{pmatrix}. \quad (69)$$

For $\Delta_g = ac - b^2 \neq 0$,

$$\mathcal{R}_{g\psi}^{\text{loc}} = -(\mathcal{A}_g^{\text{loc}})^{-1} \mathcal{B}_{g\psi}^{\text{loc}} = \frac{1}{\Delta_g} \begin{pmatrix} bv - cu \\ bu - av \end{pmatrix}. \quad (70)$$

Hence the Weyl/lensing response per unit topographic field is

$$\mathcal{R}_W = \frac{u(b - c) + v(b - a)}{2(ac - b^2)}. \quad (71)$$

The local optical metric response therefore reduces to five projected action matrix elements rather than an arbitrary transfer function.

15.3 Upstream ownership of optical covariance

Let

$$C_J = C_{\text{grav}} + \mathcal{E}_\Sigma C_\Sigma \mathcal{E}_\Sigma^\dagger + C_\xi. \quad (72)$$

With $\mathcal{G}_\psi = \mathcal{L}_\psi^{-1}$,

$$C_\psi = \mathcal{G}_\psi C_J \mathcal{G}_\psi^\dagger. \quad (73)$$

The unresolved optical forcing covariance is therefore

$$Q_{\text{opt}} = \mathcal{O}_{\text{opt}} \mathcal{R}_{g\psi} \mathcal{G}_\psi C_J \mathcal{G}_\psi^\dagger \mathcal{R}_{g\psi}^\dagger \mathcal{O}_{\text{opt}}^\dagger. \quad (74)$$

Thus Q_{opt} is not an independent phenomenological covariance matrix. Its rank and cross-observable correlations are inherited from the upstream BHSM source statistics.

For a compound-Poisson rare-event population with event rate λ_Σ and seam-source amplitude q_Σ ,

$$C_\Sigma^{\text{shot}} = \lambda_\Sigma \langle q_\Sigma q_\Sigma^\dagger \rangle, \quad (75)$$

so

$$Q_{\text{opt}}^{\text{shot}} = \lambda_\Sigma \mathcal{M}_\Sigma \langle q_\Sigma q_\Sigma^\dagger \rangle \mathcal{M}_\Sigma^\dagger, \quad \mathcal{M}_\Sigma = \mathcal{O}_{\text{opt}} \mathcal{R}_{g\psi} \mathcal{G}_\psi \mathcal{E}_\Sigma. \quad (76)$$

The remaining numerical closure is finite: evaluate the projected local metric Hessian entries (a, b, c) , the mixed entries (u, v) , the common-action seam-export normalization/orientation, and the low-order statistics of the source population. These quantities replace the previous object-by-object foreground reconstruction problem.

16 Action projection contract for the local optical Hessian

The constitutive reduction leaves five local physical coefficients (a, b, c, u, v) . These are not phenomenological parameters. Let e_A and e_ψ be the two physical scalar metric basis directions on the common closed-FLRW KKT domain. Then

$$a = \langle e_A, \hat{\mathcal{H}}_{gg} e_A \rangle, \quad (77)$$

$$b = \langle e_A, \hat{\mathcal{H}}_{gg} e_\psi \rangle, \quad (78)$$

$$c = \langle e_\psi, \hat{\mathcal{H}}_{gg} e_\psi \rangle. \quad (79)$$

Thus

$$\mathcal{A}_g^{\text{loc}} = P_g^\dagger \hat{\mathcal{H}}_{gg} P_g = \begin{pmatrix} a & b \\ b & c \end{pmatrix}. \quad (80)$$

For the normalized local topographic profile φ_{loc} , the retained mixed second variation

$$B_{g\varphi}[\gamma, \varphi] = \int d\mu_g w \gamma^{\mu\nu} \left[2F''(X)(j \cdot d\varphi) A_{\mu\nu} + 2F'(X) j_{(\mu} \partial_{\nu)} \varphi - F'(X) g_{\mu\nu} (j \cdot d\varphi) \right] \quad (81)$$

defines

$$u = B_{g\varphi}[e_A, \varphi_{\text{loc}}], \quad v = B_{g\varphi}[e_\psi, \varphi_{\text{loc}}]. \quad (82)$$

Equivalently,

$$\mathcal{B}_{g\psi}^{\text{loc}} = P_g^\dagger \hat{\mathcal{H}}_{g\varphi} \varphi_{\text{loc}} = \begin{pmatrix} u \\ v \end{pmatrix}. \quad (83)$$

If

$$c_\varphi = \langle \varphi_{\text{loc}}, \hat{\mathcal{H}}_{\varphi\varphi} \varphi_{\text{loc}} \rangle, \quad (84)$$

the same-domain Schur curvature is

$$k_{\text{loc}} = c_\varphi - (\mathcal{B}_{g\psi}^{\text{loc}})^\dagger (\mathcal{A}_g^{\text{loc}})^{-1} \mathcal{B}_{g\psi}^{\text{loc}}. \quad (85)$$

The numerical action calculation must return

$$a, b, c, u, v, c_\varphi \quad (86)$$

together with the basis/profile normalizations, $\Delta_g = ac - b^2$, the response residual $\|\mathcal{A}_g \mathcal{R}_{g\psi} + \mathcal{B}_{g\psi}\|$, the Schur curvature, the Legendre-gate status, and the common-domain status. No supernova calibration enters this extraction.

Under an invertible physical metric coordinate change $P_g \rightarrow P_g S$,

$$\mathcal{A}_g \rightarrow S^\dagger \mathcal{A}_g S, \quad \mathcal{B}_{g\psi} \rightarrow S^\dagger \mathcal{B}_{g\psi}, \quad \mathcal{R}_{g\psi} \rightarrow S^{-1} \mathcal{R}_{g\psi}, \quad (87)$$

so the reconstructed physical metric perturbation is basis invariant.

17 Spatial-current support gate for the local optical source

The retained phase-specialized metric-topographic second variation gives, on a static block-diagonal background,

$$B_{g\varphi}^{\text{spatial}}[\gamma, \varphi] = \int d\mu_g w \gamma^{ij} \left[2F''(X)(j^k \nabla_k \varphi) A_{ij} + 2F'(X) j_{(i} \nabla_{j)} \varphi - F'(X) g_{ij} (j^k \nabla_k \varphi) \right]. \quad (88)$$

Here $j_i = \langle D_i \eta, T\eta \rangle$ is the spatial target Noether current. The static conditions $D_0 \eta = 0$ and $j_0 = 0$ remove the static shift/phase block but do not force j_i to vanish.

It follows immediately that

$$j_i = 0 \quad \text{or} \quad \nabla_i \varphi = 0 \quad \implies \quad u = v = 0. \quad (89)$$

Thus a nonzero spatial target current and nonconstant local texture are necessary, though not sufficient, for this channel to source the local metric response.

For the physical scalar metric basis,

$$u = \int d\mu_g w (e_A)^{ij} \mathcal{S}_{ij}[j, \varphi], \quad (90)$$

$$v = \int d\mu_g w (e_\psi)^{ij} \mathcal{S}_{ij}[j, \varphi]. \quad (91)$$

If $\varphi = q\hat{\varphi}$ has one fixed normalized shape and stochastic amplitude q , then

$$\mathcal{B}_{g\psi}^{\text{loc}} = q b_{\text{loc}} \quad (92)$$

and

$$C_B = \text{Var}(q) b_{\text{loc}} b_{\text{loc}}^\dagger, \quad (93)$$

which has rank at most one. For m independent physical texture channels, the corresponding topographic covariance has rank at most m before additional independent source sectors are included. This gives a cross-observable low-rank prediction for the stochastic optical extension.

18 Low-rank cross-observable covariance theorem

Even before the compact full-preimage spatial-current background is numerically available, the optical construction makes a normalization-independent prediction. Let $q \in \mathbb{R}^m$ contain the active physical topographic texture amplitudes and let M_O be the complete linear response from those amplitudes to a fixed observable vector O . Then

$$C_O^{\text{topo}} = M_O C_q M_O^\dagger \quad (94)$$

and hence

$$\text{rank } C_O^{\text{topo}} \leq \text{rank } C_q \leq m. \quad (95)$$

For one active texture channel,

$$C_O^{\text{topo}} = \sigma_q^2 m_O m_O^\dagger, \quad (96)$$

so every pair satisfies

$$C_{ii} C_{jj} - |C_{ij}|^2 = 0. \quad (97)$$

Equivalently, in the noiseless isolated limit,

$$R_1 = \frac{\lambda_1}{\sum_i \lambda_i} = 1, \quad \rho_{21} = \frac{\lambda_2}{\lambda_1} = 0. \quad (98)$$

The rank statement applies to the BHSM topographic contribution, not to raw survey covariance. The test object is

$$\hat{C}_{\text{res}} = \hat{C}_{\text{obs}} - \hat{C}_{\text{std}} - \hat{N}, \quad (99)$$

with the standard-physics and measurement covariances constructed independently of the BHSM rank target.

A residual covariance requiring more than the action-allowed number of independent eigenmodes falsifies that BHSM texture-channel count even if individual pairwise correlations are nonzero. This test does not require the absolute seam-export normalization.

19 Effective representation as a compatibility audit

The role of effective-field language in the present construction is best illustrated by the curvature-aware scalar perturbation audit performed during the development of the R1 cosmological bridge. A conventional Newtonian-gauge effective representation was constructed in order to compare the geometry-first response with familiar late-time perturbation variables. After index and velocity-convention corrections, the system still exhibited a curvature-dependent global constraint-preservation defect. The corresponding flattened system closed numerically, while the curved system did not satisfy the same preservation criterion.

This result is not interpreted as a general failure of effective field theory. Nor is it used as evidence against quantum field theory. It establishes a more limited point: an effective variable set and its equations can provide a useful local or phenomenological representation without necessarily being a globally faithful realization of the complete constrained geometry.

The response of this work was therefore not to modify the effective system until it reproduced the desired cosmological signal. Instead, the defining transport was returned upstream to the physical action reduction:

$$\boxed{\text{physical Hessian} \rightarrow X_2 \rightarrow \mathcal{R}_{g2} X_2 \rightarrow \text{optical transport} \rightarrow \text{observable kernel}.} \quad (100)$$

The curvature-aware effective equations remain in the project as a compatibility audit. They are not the ontological definition of the BHSM mode.

19.1 A concrete methodological distinction

The practical difference may be summarized as follows:

Representation-first question	Geometry-first question
Which fields should be introduced?	Which perturbations survive the physical constraint and gauge quotient?
Which coefficients fit the observations?	Which coefficients are matrix elements of the retained action?
Is the effective equation stable?	Is the physical Hessian admissible on the common domain?
Can the field reproduce the signal?	Does one reduced state predict multiple observables consistently?
Can missing behavior be absorbed by new terms?	Does a failed prediction reject the proposed reduction?

This difference becomes scientifically meaningful only through prediction and failure. Several candidate reductions tested in this work fail that criterion. A homogeneous two-state mode cannot reproduce the required low-redshift localization. Preregistered foreground number-density, luminosity-weighted, and seam-weighted line-of-sight proxies do not survive held-out transfer. These failures are retained as part of the evidence because the rejected reductions were not subsequently tuned to recover the signal.

19.2 Fields as response coordinates

The geometry-first interpretation does not prohibit field variables. It changes their logical status. Once constraints, boundary data, gauge redundancy, and nonselected directions are removed, a field-like variable may serve as an efficient coordinate on the physical response. Different effective coordinate choices may therefore represent the same underlying reduced geometry.

This observation motivates the optical formulation developed in this work. A photon does not require an object-by-object effective-field inventory of everything along its path. It responds to the total spacetime geometry through the optical tidal matrix. The Jacobi/Sachs system consequently provides a direct bridge from the reduced geometry to observables without assigning fundamental status to every intermediate field representation.

20 Exact Closed-Horndeski Kernel for the First Physical Scalar Mode

The previous section uses the Bellini–Sawicki functions to organize the cubic-Galileon background. For the curvature-scale mode, however, the flat-FLRW perturbation formulas are not sufficient. We therefore specialize the non-flat Horndeski quadratic action derived by Akama and Kobayashi [6].

For

$$G_2 = X - V(T), \quad G_3 = \frac{2X}{\Lambda^3}, \quad G_4 = \frac{M_{\text{Pl}}^2}{2}, \quad G_5 = 0,$$

their tensor coefficients reduce exactly to

$$\mathcal{F}_T = \mathcal{G}_T = M_{\text{Pl}}^2.$$

Because G_4 is constant and $G_5 = 0$, the explicit curvature contributions to their Θ and Σ vanish. The remaining coefficients are

$$\Theta = HM_{\text{Pl}}^2 - \frac{\dot{T}^3}{\Lambda^3}$$

and

$$\Sigma = \frac{1}{2}\dot{T}^2 + \frac{12H\dot{T}^3}{\Lambda^3} - 3M_{\text{Pl}}^2H^2.$$

Equivalently,

$$\Theta = HM_{\text{Pl}}^2 \left(1 - \frac{\alpha_B}{2}\right).$$

Define

$$r(t) \equiv \frac{M_{\text{Pl}}^2 \Sigma}{\Theta^2}, \quad \kappa(t) \equiv \frac{K}{a^2}.$$

The first physical scalar harmonic has alternative-convention index $N = 3$, so

$$k^2 = K(N^2 - 1) = 8K.$$

The two Akama–Kobayashi scalar constraints then reduce to

$$\Sigma \delta n + 8\Theta\kappa\chi + 3\Theta\dot{\zeta} + 5M_{\text{Pl}}^2\kappa\zeta = 0,$$

$$\Theta\delta n - M_{\text{Pl}}^2\dot{\zeta} - M_{\text{Pl}}^2\kappa\chi = 0.$$

They can be solved algebraically:

$$\delta n = \frac{5M_{\text{Pl}}^2}{\Theta(r+8)} \left[\dot{\zeta} - \frac{M_{\text{Pl}}^2\kappa}{\Theta}\zeta \right]$$

and

$$\chi = -\frac{r+3}{\kappa(r+8)}\dot{\zeta} - \frac{5M_{\text{Pl}}^2}{\Theta(r+8)}\zeta.$$

The reduced kinetic coefficient of this $n = 2$ mode is therefore

$$\mathcal{G}_{S,2} = M_{\text{Pl}}^2 \frac{5(r+3)}{r+8}.$$

Using the cubic-Galileon α -functions gives the exact identity

$$r+3 = \frac{D_{\text{kin}}}{2(1-\alpha_B/2)^2},$$

so that

$$\mathcal{G}_{S,2} = M_{\text{Pl}}^2 \frac{5D_{\text{kin}}}{D_{\text{kin}} + 10(1-\alpha_B/2)^2}.$$

Thus, provided $\Theta \neq 0$, positivity of the previously computed D_{kin} is sufficient for positivity of the pure gravity–scalar kinetic coefficient of the first physical closed-universe scalar harmonic.

For completeness, the corresponding non-flat Horndeski gradient coefficient is

$$\mathcal{F}_{S,2} = \frac{1}{a} \frac{d}{dt} \left[\frac{5}{r+8} \left(\frac{aM_{\text{Pl}}^4}{\Theta} \right) \right] - M_{\text{Pl}}^2 + \frac{5}{r+8} \frac{M_{\text{Pl}}^6}{\Theta^2} \frac{K}{a^2}.$$

The repository evaluates this expression as a gravity–scalar diagnostic only. The late-time R1 background contains matter and radiation, whose perturbations introduce additional dynamical variables and source terms. Consequently, $\mathcal{F}_{S,2}$ is not promoted here to a complete matter-era stability or light-cone transfer result.

This section closes an important part of the bridge: the first physical harmonic has an explicit constrained scalar–metric kernel rather than only a geometric identification. The remaining observable calculation is the coupled matter/radiation plus topographic-response system and its gauge-invariant transformation to the metric and velocity variables entering distance, growth, and lensing observables.

21 Matter-Sourced Constraint Transfer at n=2

The Round-3 kernel established the pure gravity–scalar constrained dynamics of the first physical closed-universe harmonic. The next step is to introduce minimally coupled matter sources without

importing a flat-space or quasi-static approximation. We use the curvature-aware Newtonian-gauge EFT equations written by Gambino and Pace [7]. Their metric convention is

$$ds^2 = -(1 + 2\Phi)dt^2 + a^2(1 - 2\Psi)\gamma_{ij}dx^i dx^j,$$

with

$$\delta T^0_0 = -\delta\rho, \quad \delta T^0_i = (\rho + p)D_i v.$$

For the present cubic-Galileon subclass,

$$\alpha_M = \alpha_T = \alpha_H = 0, \quad M^2 = M_{\text{Pl}}^2.$$

The curvature-aware equations use the Gleyzes/Gambino–Pace braiding normalization. To distinguish it from the Bellini–Sawicki braiding function used in the previous sections, we write

$$\boxed{\alpha_B^{\text{GP}} = -\frac{1}{2}\alpha_B^{\text{BS}}}.$$

The kineticity is unchanged,

$$\alpha_K^{\text{GP}} = \alpha_K^{\text{BS}},$$

and therefore

$$\alpha_K + 6(\alpha_B^{\text{GP}})^2 = \alpha_K + \frac{3}{2}(\alpha_B^{\text{BS}})^2 = D_{\text{kin}}.$$

For ideal dust and ideal radiation we set the scalar anisotropic stress to zero, so the traceless spatial equation gives

$$\boxed{\Phi = \Psi}.$$

On the physical $n = 2$ harmonic,

$$D^2 Q = -8KQ, \quad (D^2 + 3K)Q = -5KQ.$$

Writing

$$\kappa = \frac{K}{a^2},$$

the sourced 00 equation becomes

$$\begin{aligned} & 6(1 + \alpha_B^{\text{GP}})H\dot{\Phi} + (\alpha_K - 6\alpha_B^{\text{GP}})H^2\dot{\pi} + \left[(6 - \alpha_K + 12\alpha_B^{\text{GP}})H^2 + 10\kappa\right]\Phi \\ & + 6H\left[\frac{\rho + p}{2M_{\text{Pl}}^2} + \dot{H}(1 + \alpha_B^{\text{GP}}) + \kappa\left(1 - \frac{8}{3}\alpha_B^{\text{GP}}\right)\right]\pi = -\frac{\delta\rho}{M_{\text{Pl}}^2}. \end{aligned}$$

The $0i$ equation is

$$2\dot{\Phi} - 2H\alpha_B^{\text{GP}}\dot{\pi} + 2(1 + \alpha_B^{\text{GP}})H\Phi + \left(2\dot{H} + \frac{\rho + p}{M_{\text{Pl}}^2}\right)\pi = -\frac{q}{M_{\text{Pl}}^2},$$

where

$$q = \rho_m v_m + \frac{4}{3}\rho_r v_r, \quad \delta\rho = \rho_m \delta_m + \rho_r \delta_r.$$

These equations form a 2×2 local constraint-transfer system for

$$(\dot{\Phi}, \dot{\pi}).$$

Its determinant is

$$\boxed{\det \mathcal{C} = -2H^2 \left[\alpha_K + 6(\alpha_B^{\text{GP}})^2 \right] = \boxed{-2H^2 D_{\text{kin}}}.}$$

Thus the same positive- D_{kin} branch used in the background and Round-3 kinetic audit guarantees that this sourced rate block is nonsingular.

For the velocity convention above, covariant conservation gives

$$\dot{\delta}_m = \frac{k^2}{a^2} v_m + 3\dot{\Phi}, \quad \dot{v}_m = -\Phi,$$

and

$$\dot{\delta}_r = \frac{4}{3} \frac{k^2}{a^2} v_r + 4\dot{\Phi}, \quad \dot{v}_r = -\Phi - \frac{1}{4} \delta_r,$$

with

$$\frac{k^2}{a^2} = 8\kappa.$$

The repository implements these equations as a local linear transfer matrix with the additional topographic response explicitly switched off. This is deliberately not yet promoted to a globally integrated observable solution. The curvature-aware EFT equations and the Round-3 ADM kernel use different perturbation variables and normalizations; the remaining evolution equations must first be matched analytically and then subjected to a Bianchi/constraint-preservation audit. Only after that compatibility gate is passed will we integrate the full matter-era $n = 2$ system and add the independent reduced topographic operator.

22 ADM–EFT Dictionary and Differential-Algebraic Consistency

Rounds 3 and 4 use complementary perturbation languages. Akama and Kobayashi formulate the non-flat Horndeski action in unitary ADM gauge,

$$N = 1 + \delta n, \quad N_i = D_i \chi, \quad g_{ij} = a^2 e^{2\zeta} \gamma_{ij},$$

while Gambino and Pace restore the scalar with

$$t \rightarrow t + \pi$$

and give the curvature-aware field equations in Newtonian gauge [6, 7].

The linear time shift that removes the scalar ADM shift gives

$$\boxed{\pi = -\chi},$$

$$\boxed{\Phi = \delta n - \dot{\pi} = \delta n + \dot{\chi}},$$

and

$$\boxed{\Psi = -\zeta + H\pi = -\zeta - H\chi}.$$

The inverse dictionary is

$$\boxed{\chi = -\pi, \quad \delta n = \Phi + \dot{\pi}, \quad \zeta = H\pi - \Psi.}$$

22.1 Correct Bianchi audit

The first implementation attempt treated the 00 and 0*i* equations as a complete local evolution system and then demanded that the spatial-trace and scalar equations vanish for an arbitrary pointwise state. That test is too strong and uses the constraint equations in the wrong role.

The correct formulation is differential algebraic. The spatial-trace and scalar equations contain the second derivatives of the physical scalar sector and determine

$$(\ddot{\Phi}, \ddot{\pi}),$$

while the 00 and 0*i* equations define constraints on the initial data. Covariant consistency requires those constraints to be preserved by the evolution.

For

$$\alpha_M = \alpha_T = \alpha_H = 0, \quad \Phi = \Psi,$$

the acceleration matrix is

$$\mathcal{A} = \begin{pmatrix} 2 & -2H\alpha_B \\ 6H\alpha_B & H^2\alpha_K \end{pmatrix},$$

where α_B is in the Gambino–Pace/Gleyzes normalization. Its determinant is

$$\det \mathcal{A} = 2H^2 (\alpha_K + 6\alpha_B^2) = 2H^2 D_{\text{kin}} > 0.$$

Thus the second-order scalar–metric evolution block is nonsingular on the same positive- D_{kin} branch.

The repository now integrates the trace/scalar equations together with ideal dust and radiation conservation, starting from data that satisfy the sourced 00 and 0*i* constraints. The Bianchi audit is then the preservation of those two constraints over the interval

$$2.1 \geq z \geq 0.$$

Both raw and term-normalized constraint residuals are evaluated. The topographic reduced-response operator remains switched off throughout this certification.

23 Selective Harmonic Response

Let

$$L_c(a) = aR_H\ell, \quad L_c^2 p_n = \ell^2 n(n+2),$$

and write

$$K_n = p_n C_n, \quad C_n = c_s^2 - \chi \left[1 - \ell^2 n(n+2) \right].$$

The physical scalar spectrum begins at $n = 2$. Selective softening of $n = 2$ with stiffening of every propagating $n \geq 3$ requires

$$1 - 8\ell^2 > 0, \quad 1 - 15\ell^2 < 0,$$

so

$$\frac{1}{15} < \ell^2 < \frac{1}{8}.$$

Then

$$C_2 < c_s^2, \quad C_n > c_s^2 \quad (n \geq 3).$$

The formal $n = 1$ coefficient also lies on the soft side, but $n = 1$ is not a propagating scalar degree of freedom in the regular one-field closed-FLRW sector.

If $0 < C_2 \ll 1$,

$$T_2 \simeq \frac{S_2}{p_2 C_2}$$

can be enhanced while higher physical harmonics remain regular. For the next physical level,

$$\frac{T_3}{T_2} = \frac{S_3}{S_2} \frac{8C_2}{15C_3},$$

so

$$\left| \frac{T_3/S_3}{T_2/S_2} \right| = \frac{8C_2}{15C_3} < \frac{8}{15}.$$

For the pure-dipole A_{4i} subspace of the selected $n = 2$ harmonic, the observer-sky pattern is exactly dipolar with geometric radial envelope $\sin 2\psi$.

24 Observable Projection

The observable distance perturbation is not identified directly with the scalar amplitude. At linear order it is a light-cone projection of metric, velocity, and scalar perturbations:

$$\frac{\delta D_L}{\bar{D}_L} = \int d\chi \left[W_\Phi \Phi + W_\Psi \Psi + W_v v_\parallel + \dots \right].$$

If the selected $n = 2$ curvature-scale sector admits a physical gauge-invariant metric response, its exact scalar angular dependence can factor as

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{p}},$$

leading schematically to

$$\frac{\delta D_L}{\bar{D}_L} = \epsilon_D(z) \hat{\mathbf{n}} \cdot \hat{\mathbf{p}}.$$

The existence and normalization of $\epsilon_D(z)$ are therefore bridge outputs to be derived, not consequences of the scalar harmonic geometry alone.

For supernovae,

$$\mu = 5 \log_{10} D_L + \text{const},$$

and therefore a small magnitude dipole A_μ corresponds to

$$\frac{\delta D_L}{\bar{D}_L} = \frac{\ln 10}{5} A_\mu \cos \theta.$$

For structure growth we define

$$k^2 \Psi = -4\pi G a^2 \mu(k, a) \rho_m \delta_m, \quad \Phi = \gamma(k, a) \Psi,$$

and

$$\Sigma(k, a) = \frac{\mu(k, a)}{2} [1 + \gamma(k, a)].$$

The matter growth factor obeys

$$D'' + \left(2 + \frac{H'}{H} \right) D' - \frac{3}{2} \Omega_m(a) \mu(k, a) D = 0,$$

with prime denoting $d/d \ln a$. The observable is

$$f\sigma_8(k, z) = \frac{D'}{D} D \sigma_{8,\text{ini}} = D'(k, z) \sigma_{8,\text{ini}}.$$

The quasi-static limit is used only on scales inside the scalar sound horizon [8]. Horizon-scale $n = 1$ observables require the full constrained long-wavelength response and must not be obtained by extrapolating quasi-static formulas into the first physical curvature-scale mode.

25 Predictions and Falsification Criteria

25.1 Low-redshift calibration

The Pantheon+ data release and cosmological analysis provide the supernova basis for the carried-forward tomographic calibration used here [9, 10]. The earlier scalar-topographic analysis reported

$$A_\mu^* = -0.0412 \pm 0.009$$

in the bin

$$0.01 < z < 0.03,$$

with best-fit direction

$$(\text{RA}, \text{Dec}) = (211.48^\circ, -12.81^\circ).$$

This corresponds to

$$\frac{\delta D_L}{D_L} = (-0.01895 \pm 0.00414) \cos \theta,$$

and at sufficiently low redshift,

$$\frac{\delta H}{H} = (+0.01895 \pm 0.00414) \cos \theta.$$

These numbers are a carried-forward calibration input; the present repository does not independently rerun the Pantheon+ tomographic likelihood. They normalize only the observed $n = 1$ response,

$$A_\mu = \frac{\mathcal{K}_{\text{SN}} S_2}{p_2 C_2},$$

and do not independently determine S_2 , $\mathcal{K}_{\text{SN}}$, or C_2 . Their interpretation as a metric/light-cone response is additionally conditional on resolving the exceptional closed-FLRW $n = 1$ sector.

25.2 Curvature

The pre-existing prospective curvature forecast is retained without retuning:

$$\boxed{\Omega_k = -0.018 \pm 0.004}.$$

At $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$,

$$R_H = \frac{c}{H_0 \sqrt{|\Omega_k|}} \simeq 33.15 \text{ Gpc}.$$

The sign prediction is $\Omega_k < 0$ in the usual FLRW curvature convention.

This value is a frozen model forecast, not a current empirical best fit. DESI DR2 BAO results are reported as well described by flat Λ CDM, even while combinations with other probes motivate tests of extensions such as dynamical dark energy [11]. The purpose of retaining -0.018 ± 0.004 is therefore prospective falsification: it must be compared against independent curvature inference without being moved toward the observed result.

25.3 Harmonic hierarchy

The bridge predicts selective $n = 2$ softening with no analogous $n \geq 3$ enhancement. A future source-normalized S^3 $n = 2$ susceptibility comparable to or larger than the $n = 1$ susceptibility would reject the minimal harmonic-filter branch. This statement is made at the compact-space harmonic level; comparison to a measured sky quadrupole requires an explicit observer-shell projection.

25.4 Fixed-axis test

A prospective independent supernova sample should first test the carried-forward axis

$$(211.48^\circ, -12.81^\circ)$$

rather than scanning freely over the sky. A free-axis search may be reported secondarily.

25.5 High- n recovery

At large n ,

$$C_n \simeq \chi \ell^2 n^2,$$

so the extra scalar susceptibility falls as n^{-2} . The model therefore predicts a strong separation between the first nonconstant compact-space level and conventional clustering modes.

25.6 Failure conditions

The minimal bridge is disfavored if one or more of the following are established with adequate precision:

1. independent curvature measurements exclude the frozen negative- Ω_k forecast;
2. the source-normalized S^3 $n = 2$ response is comparable to or exceeds $n = 1$;
3. an independent low-redshift sample excludes the frozen dipole axis while finding a stable incompatible physical axis;
4. a coherent metric/light-cone dipole of comparable amplitude persists into a redshift regime in which the completed model predicts loss of enhanced susceptibility;
5. ordinary RSD modes show a large scale-independent modification incompatible with the high- n suppression;
6. lensing requires a metric response incompatible with the same completed scalar-metric propagator that controls matter clustering;
7. no gauge-invariant physical metric response exists for the exceptional closed-FLRW harmonic corresponding to the proposed $n = 1$ topographic mode.

26 Implications of the geometry-first workflow

The significance of the present construction is not that one more scalar cosmology has been added to the literature. The calculation demonstrates a different route from microscopic assumptions to observations.

At the BHSM level, the starting object is a constrained geometric action. Physical perturbations are selected only after the constraint surface, boundary conditions, gauge quotient, and physical Hessian are specified. At the cosmological level, the same methodology selects a physical compact topographic carrier, induces its metric response, transports light through the resulting geometry, and ends with observer-frame kernels.

The resulting chain is

$$\boxed{\text{geometry} \rightarrow \text{enclosure} / \text{physical reduction} \rightarrow \text{response} \rightarrow \text{optical propagation} \rightarrow \text{observation.}} \quad (101)$$

This viewpoint also clarifies the relation between BHSM and cosmology. The cosmological sector is not an independent effective theory appended to BHSM. It is a large-scale test of whether the same reduction logic can generate observable response after the physical mode count is fixed.

26.1 Predictive content

Several predictions or null conditions arise from the reduction itself:

1. the exceptional lowest scalar harmonic is not promoted to the ordinary propagating scalar carrier;
2. the first physical scalar carrier is the $n = 2$ branch used in the observable transfer;
3. one scalar supernova calibration has rank one on a two-dimensional physical state and cannot determine the remaining state direction;
4. the pure one-mode response has a common leading observer-sky axis across linear observables;
5. a model with m independent physical stochastic texture channels has topographic covariance rank at most m after standard covariance and measurement noise are accounted for;
6. failed homogeneous and foreground reductions are rejected rather than repaired by post-hoc parameter changes.

These statements differ in maturity and claim status, but they share a common feature: they constrain future observations rather than merely describing existing data.

26.2 What remains open

The present paper does not claim a complete microscopic BHSM derivation of the cosmological source normalization. Important upstream quantities remain action-level open, including the normalized compact full-preimage background, the common-domain local metric/topographic matrix elements, the exact seam-to-topography export normalization, and the statistical law of rare source events.

The geometry-first methodology makes these absences explicit. They are not replaced by fitted optical coefficients. The photon/observable propagation can therefore be used while the remaining upstream source calculation continues.

26.3 Scientific criterion

The proposed workflow succeeds only if the same reduced physical structure predicts multiple observables more economically and consistently than representation-specific alternatives. It should be rejected if future data require incompatible axes, incompatible redshift transfer, excessive cross-observable covariance rank, or additional independent response channels not supplied by the retained action.

The purpose of geometry-first modeling is therefore not to avoid falsification. It is to move the point of falsification closer to the underlying physical assumptions.

27 Numerical Implementation and Preregistration Protocol

The numerical pipeline is separated into

$$\boxed{\text{background} \rightarrow \text{physical scalar response} \rightarrow \text{observables} \rightarrow \text{comparison}.}$$

Prediction generation and data comparison are implemented in separate scripts.

27.1 Background system

Let $v = \dot{T}$. Expanding the scalar current equation gives

$$\left(1 + \frac{12Hv}{\Lambda^3}\right) \dot{v} + \frac{6v^2}{\Lambda^3} \dot{H} = \mu^3 - 3H \left(v + \frac{6Hv^2}{\Lambda^3}\right).$$

The Raychaudhuri equation gives

$$\frac{2v^2}{\Lambda^3} \dot{v} - 2M_{\text{Pl}}^2 \dot{H} = \rho_m + \frac{4}{3}\rho_r + v^2 + \frac{6Hv^3}{\Lambda^3} - \frac{2M_{\text{Pl}}^2}{a^2 R_H^2}.$$

At each integration step these equations form a 2×2 linear system for $(\dot{v}, \dot{H})$. The remaining equations are

$$\dot{T} = v, \quad \dot{a} = aH.$$

The Friedmann equation is retained as an independent residual check:

$$3M_{\text{Pl}}^2 \left(H^2 + \frac{1}{a^2 R_H^2} \right) - (\rho_m + \rho_r + \rho_T) = 0.$$

27.2 Dimensionless variables

We integrate in $N = \ln a$ with

$$E = \frac{H}{H_0}, \quad \phi = \frac{T}{M_{\text{Pl}}}, \quad \lambda = \frac{\Lambda^3}{M_{\text{Pl}} H_0^2}, \quad q = \frac{\mu^3}{M_{\text{Pl}} H_0^2}.$$

The exploratory reference branch R1 uses

$$\Omega_{m0} = 0.31, \quad \Omega_{r0} = 9.0 \times 10^{-5}, \quad \Omega_{k0} = -0.018, \quad \lambda = 1, \quad q = 3.$$

The constant V_0 is determined only by the shooting condition $E(a = 1) = 1$.

27.3 Initial conditions

The numerical integration begins at

$$z_i = 100.$$

The decaying homogeneous current solution is removed using the matter-era particular solution

$$J_i \simeq \frac{2\mu^3}{9H_i}.$$

The initial scalar velocity is the positive small-velocity root of

$$v_i + \frac{6H_i v_i^2}{\Lambda^3} = J_i.$$

27.4 Growth grid

Before prospective comparison, predictions are generated on the fixed grid

$$k = \{0.02, 0.05, 0.10, 0.15, 0.20\} \text{ Mpc}^{-1}$$

and

$$z = \{0.5, 0.8, 1.0, 1.5, 2.1\}.$$

A deterministic JSON manifest containing the action version, parameters, numerical tolerances, calibration values, and predicted observables is hashed with SHA-256. Any later scientific or numerical revision creates a new manifest version rather than overwriting the previous freeze.

28 Prospective observational test matrix

The geometry-first workflow becomes scientifically useful only through prospective tests that couple more than one observable to the same reduced physical state. Table 1 summarizes the principal tests without introducing a new fit in this section.

The one-channel rank test has an amplitude-independent null condition,

$$C_{ii}C_{jj} - |C_{ij}|^2 = 0, \tag{102}$$

for every 2×2 minor of an isolated rank-one topographic covariance. Standard astrophysical covariance and measurement noise must be modeled or subtracted before this condition is assessed.

The numerical normalization of every row in Table 1 is not claimed closed at present. The purpose of the table is to specify which pieces of future data must be explained by one common physical response rather than by separate per-observable fits.

29 Results: Reference Branch R1

R1 is an exploratory reference branch rather than a uniquely derived BHSM cosmology. Its purpose is to demonstrate that the bridge can simultaneously realize late-time scalar activation, a selectively enhanced $n = 1$ response, and near-standard high- n structure growth.

For

$$\Omega_{m0} = 0.31, \quad \Omega_{r0} = 9.0 \times 10^{-5}, \quad \Omega_{k0} = -0.018, \quad \lambda = 1, \quad q = 3,$$

Observable	Theory object	Prospective signature	Failure mode
SN / standard candles	$F_{D_L}(z)X_2$	Frozen redshift-dependent distance response with common angular structure	Amplitude or redshift evolution inconsistent with the independently selected state
Growth / RSD	$F_{f\sigma_8}(z)X_2$	Same physical state propagated into matter growth	Requires a second independent state not present in the reduction
Weak lensing / Weyl	$F_W(z)X_2$ and optical shear	Cross-observable angular and redshift consistency	Axis or transfer incompatible with the distance response
Photon redshift / convergence / shear	Y_γ	Joint optical-state covariance generated by common geometry	Observed covariance structure incompatible with propagated optical state
Residual covariance	$C_{\text{topo}} = M_O C_q M_O^\dagger$	$\text{rank } C_{\text{topo}} \leq m$	Residual rank robustly exceeds the physically admitted channel count

Table 1: Prospective tests of the geometry-first cosmological response. The tests constrain a common reduced state and common source architecture rather than fitting each observable independently.

the shooting condition $E(1) = 1$ gives

$$\frac{V_0}{M_{\text{Pl}}^2 H_0^2} = 2.39861.$$

The scalar braiding and no-ghost combination evolve as shown in Table 2.

z	α_B	D_{kin}	Ω_T
10	2.07×10^{-7}	1.64×10^{-6}	0.00194
3	8.60×10^{-5}	6.80×10^{-4}	0.03915
2.1	3.72×10^{-4}	0.00294	0.08060
1.5	0.00122	0.00959	0.14321
1.0	0.00380	0.02988	0.24551
0.8	0.00622	0.04889	0.30787
0.5	0.01346	0.10536	0.43200
0.3	0.02271	0.17740	0.53538
0	0.04940	0.38485	0.70791

Table 2: Reference-branch background evolution.

The scalar equation of state evolves from nearly vacuum-like behavior at high redshift to

$$w_T(z=0) \simeq -0.906,$$

and the background acceleration transition occurs at

$$z_{\text{acc}} \simeq 0.683.$$

Relative to a curved Λ CDM model with identical $H_0, \Omega_{m0}, \Omega_{r0}, \Omega_{k0}$, the R1 expansion rate differs by only order one percent over the late-time interval. The maximum difference in the tabulated range is about 1.7% near $z \sim 0.5$.

29.1 Prospective growth target

In the minimal high- n realization, the direct topographic fifth-force term is negligible on ordinary RSD modes and the dominant growth shift is produced by the altered background history. With identical matter-era normalization,

$$R_{f\sigma_8}(z) = \frac{f\sigma_{8,\text{R1}}}{f\sigma_{8,\Lambda\text{CDM}}}$$

is

The central phenomenological result is therefore not a large generic modification of gravity. It is the combination

enhanced horizon-scale $n = 1$ susceptibility
 + rapid high- n recovery
 + few-percent late-time growth shift

z	$R_{f\sigma_8}$	suppression
2.1	0.99346	0.65%
1.5	0.98869	1.13%
1.0	0.98164	1.84%
0.8	0.97795	2.20%
0.5	0.97271	2.73%
0.3	0.97168	2.83%
0.0	0.98482	1.52%

Table 3: R1 growth forecast relative to matched curved Λ CDM. These values are reference-branch predictions and become formally preregistered only when reproduced by the canonical code and written to the hashed manifest.

30 Reference-Branch Figures

31 Discussion

The closed-FLRW gauge audit produces a concrete correction. The first nonconstant scalar harmonic in our original notation, $n = 1$, maps to the exceptional $N = 2$ sector of standard closed-universe perturbation theory. Its scalar-derived trace-free tensor harmonic vanishes, and the regular generally covariant one-field scalar sector contains no propagating scalar degree of freedom there. Modern closed-inflation calculations likewise take physical scalar modes to begin at $N \geq 3$ [2, 3].

The natural response is to move the topographic carrier to the first physical scalar level: our $n = 2$, equivalent to $N = 3$. A degree-two harmonic on S^3 is a traceless quadratic form in the $\mathbb{R}^4$ embedding coordinates. Its cross components A_{4i} generate an exact observer-sky dipole

$$T_{2,\text{dip}} \propto \sin 2\psi (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}})$$

without requiring a sky quadrupole.

The reduced-response filter also adapts naturally. The corrected window

$$\frac{1}{15} < \ell^2 < \frac{1}{8}$$

softens $n = 2$ while stiffening every higher propagating scalar harmonic $n \geq 3$. The formal $n = 1$ coefficient is irrelevant to the regular physical scalar spectrum. The enhanced carrier is therefore the first physical compact-space scalar mode.

The role of BHSM remains structural rather than a completed microscopic derivation. The imported element is the constraint-reduced response logic. The specific mixed block and the source law S_2 remain exploratory until derived from the parent boundary action.

The complete $n = 2$ scalar-metric transfer kernel must still be derived in the closed cubic-Galileon background. $D_{\text{kin}} > 0$ remains an inherited kinetic diagnostic rather than a complete closed-background stability proof. R1 values $q = 3$ and $\lambda = 1$ remain reference choices, and the frozen curvature forecast must not be retuned to observational results.

The R1 numerical background and high- n growth forecast do not depend on this harmonic correction. Their frozen manifest is therefore preserved unchanged, while the theoretical amendment is recorded separately before comparison with the targeted independent datasets.

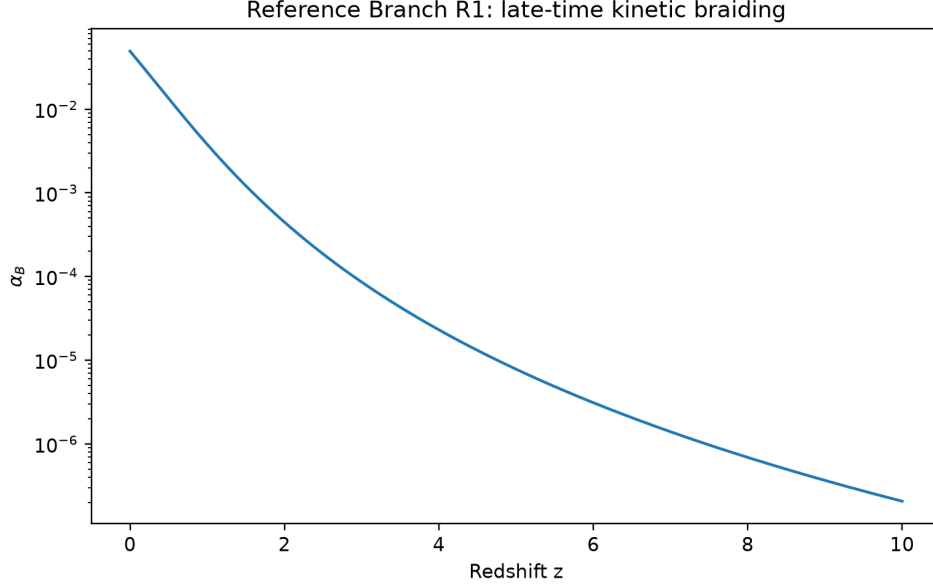


Figure 1: Kinetic-braiding function $\alpha_B(z)$ for Reference Branch R1. The logarithmic vertical scale emphasizes the suppression of the braiding sector toward high redshift.

32 Conclusions

The closed-FLRW gauge analysis resolves the principal ambiguity in the original bridge construction. Our $n = 1$ scalar harmonic maps to the exceptional $N = 2$ sector and is not retained as the ordinary propagating scalar carrier. The first physical scalar level is

$$\boxed{n_\star = 2} \quad (\boxed{N_\star = 3}).$$

This first physical harmonic contains an exact sky-dipole subspace,

$$T_{2,\text{dip}} = A \sin[2\psi(z)](\hat{\mathbf{n}} \cdot \hat{\mathbf{p}}),$$

when only the cross components A_{4i} are populated.

The corrected selective-response window is

$$\boxed{\frac{1}{15} < \ell^2 < \frac{1}{8}},$$

within which $n = 2$ is softened and every higher propagating mode $n \geq 3$ is stiffened. The next-level source-normalized response obeys

$$\left| \frac{T_3/S_3}{T_2/S_2} \right| = \frac{8C_2}{15C_3} < \frac{8}{15}.$$

The carried-forward supernova amplitude now calibrates the physical $n = 2$ dipole response. The R1 background and high- n growth predictions remain unchanged.

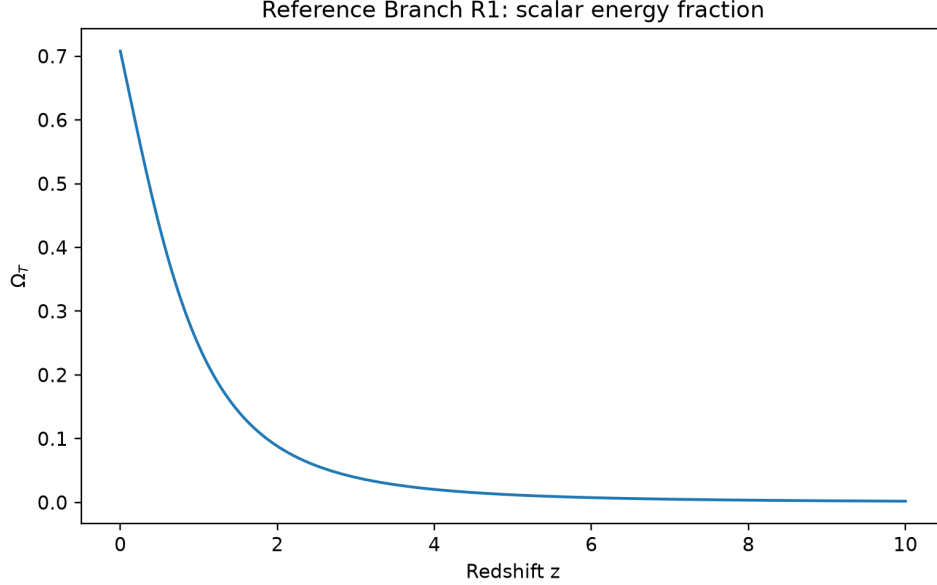


Figure 2: Scalar energy fraction $\Omega_T(z)$ for R1. The scalar component is subdominant at high redshift and becomes dynamically important only at late times.

The revised signature is therefore

- first physical S^3 scalar mode ($n = 2$) enhanced
- $\oplus$ fixed-axis dipole subspace with $\sin 2\psi$ envelope
- $\oplus$ suppressed $n \geq 3$ susceptibility
- $\oplus$ near-GR high- k growth.

The next calculation is the full gauge-invariant $n = 2$ scalar–metric transfer matrix in the closed cubic-Galileon background.

33 Conclusion

This work develops a geometry-first workflow in which the physical degrees of freedom are outputs of constrained reduction rather than inputs chosen as fundamental fields. BHSM supplies the organizing principle: vary first, enforce constraints and boundary data, quotient gauge redundancy, construct the physical Hessian, and only then identify the response variables that propagate to observables.

Applied to closed hyperspherical cosmology, that procedure yields a physical $n = 2$ reduced state, an induced metric response, a gauge-invariant luminosity-distance kernel, and a Jacobi/Sachs optical transport system for redshift, convergence, and shear. Unresolved topographic structure can be propagated statistically, leading to a low-rank cross-observable covariance prediction when only a small number of physical texture channels are active.

Effective-field descriptions remain valuable comparison tools, but they are not taken here as the defining ontology of the construction. The curvature-dependent compatibility failure encountered in one effective representation and the rejection of several preregistered foreground models illustrate the practical importance of keeping representation distinct from physical reduction.

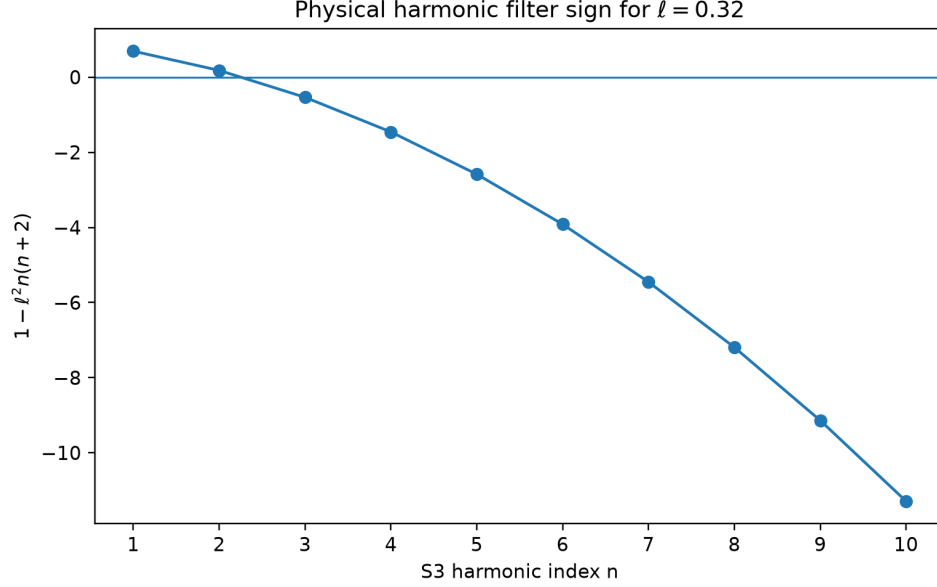


Figure 3: Sign of the Schur-response factor $1 - \ell^2 n(n+2)$ for the illustrative filtering value $\ell = 0.40$. The $n = 1$ branch lies on the softening side while every displayed $n \geq 2$ mode lies on the stiffening side.

The remaining challenge is upstream rather than observational: derive the common-action normalization and physical source statistics of the local topographic response. The downstream chain from physical response to photon transport and observable kernels is now sufficiently explicit to make those future source calculations empirically consequential.

The central proposal is therefore methodological and falsifiable: geometry and physical reduction should precede field representation, and the surviving reduced structure must earn its status by predicting several observables with one common response architecture.

Data Availability Statement

The observational inputs used in this work are drawn from publicly available astronomical data products identified in the corresponding analysis sections. The derived code, frozen manifests, machine-readable artifacts, and regression tests supporting the reported calculations are maintained in the project repository. A permanent archival release and DOI should be supplied with the submitted version so that the exact manuscript checkpoint can be reproduced.

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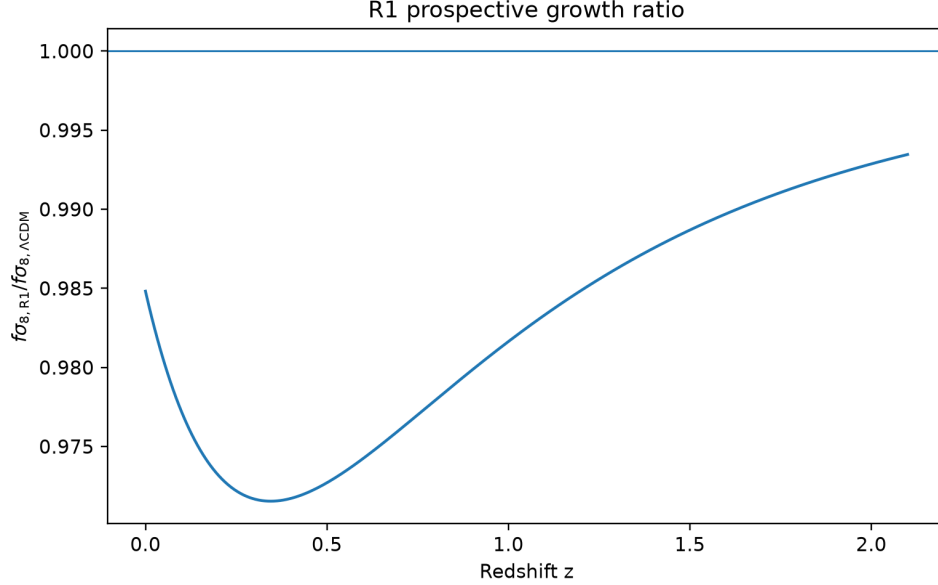


Figure 4: Prospective R1 growth ratio relative to a matched curved- Λ CDM background with the same early normalization.

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